

LLM-Powered Multi-Agent Systems: A Survey of Collaboration and Learning Strategies

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Abstract

With the progress in Large Language Models (LLMs), research in this area has grown rapidly. This has led to the development of LLM-based Multi-Agent Systems (MAS), which use multiple advanced models trained on large amounts of data. The main goal is to mimic human-like teamwork, allowing different agents to work together to solve complex problems. By splitting tasks among specialized agents, these systems can overcome challenges that a single LLM agent might struggle with, making them more efficient and effective. In this paper, we provide a comprehensive review of LLM based MAS, offering a systematic analysis of their frameworks. First, we explore the construction of MAS, introducing a categorization of collaboration structures based on existing research. Second, we examine the communication mechanisms within MAS, which define the strategies agents use to exchange information and make decisions. Next, we review the learning methods and evaluation metrics employed in MAS, highlighting how agents adapt their responses and optimize their performance for improved results, and how we can evaluate them based on the domain, and whether it is single or multi-agents' structure. Finally, to conclude this review on LLM based MAS, we propose a comprehensive road map (Figure 4), intended to support researchers and practitioners in efficiently designing and implementing multi-agent solutions tailored to domain applications.

CCS Concepts

• **Information systems—Information systems applications—Multimedia information systems;**

Keywords

Large Language Model, Multi- Agent Collaboration, Artificial Intelligence, Survey

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1 INTRODUCTION

Large Language Models (LLMs) have demonstrated human-like decision-making and intelligence [9], [76], [2], [97], [90], [85], [47]. Trained in vast amounts of publicly available data, LLMs are equipped to perform sophisticated tasks, including reasoning, creative writing, and facilitating interactions [85], [104]. The natural language interface provided by LLMs makes these interactions more flexible and explainable for humans. Due to these capabilities, many research projects are now focused on leveraging LLMs across various domains to complete tasks more effectively [121], [82], [75]. To address some of the challenges associated with LLMs, researchers are tackling issues such as hallucinations [81], [68], bias and fairness [105], [71], computational cost and energy consumption [51], and context length limitations [21], [30].

LLM based MAS defines LLMs that collaborate with each other as a team of collective intelligence to share their knowledge and solve problems. The ability of MAS to facilitate knowledge sharing among multiple agents while enabling specialization in distinct roles makes it highly effective for solving complex problems in various domains, including software development, game design, and disease propagation modeling [106]. This mirrors the dynamics of human society, where individuals collaborate as a team, each assuming a specialized role to collectively achieve a common goal [43]. A key focus in MAS is the collaboration paradigm and communication structures that facilitate agent interaction, enabling them to connect, share plans, and make collective decisions [56] [72]. Another crucial aspect is how these agents learn and adapt through teamwork, improving their performance in ways that may align with or differ from traditional machine learning approaches. With the rapid growth of this field, a comprehensive systematic review of the collaboration paradigms and learning methods in MAS is essential to provide researchers with a structured overview and serve as a blueprint for future studies. Although existing surveys focus on other aspects of MAS, this paper specifically examines these critical dimensions to fill the literature gap. The main contributions of this research are listed below.

- **Comprehensive Analysis of MAS Collaboration Structures:** This paper provides a systematic review of different structural frameworks used in MAS, including centralized, decentralized, and hybrid collaboration models, and examines how communication networks influence agent interaction and decision-making.

- **Evaluation of Multi-Agent Collaboration Strategies:** The study explores various strategies agents use to work together, such as competition and debate and discusses the effectiveness of these strategies in different domains, including robotics and game development.
- **Investigation of Multi-Agent Learning Strategies:** The paper reviews learning mechanisms within MAS, including reinforcement learning, prompt-based learning and highlights how agents adapt, optimize decision-making, and improve cooperation over time.
- **Analysis of Evaluation Metrics Across Domains:** The paper outlines key evaluation metrics used to assess the performance of both single-agent and MASs across major domains.

2 MULTI- AGENTS COLLABORATION STRUCTURE

In LLM-based MASs, the collaboration structure determines how agents interact to achieve their assigned objectives. It serves as the underlying architecture that facilitates communication and information exchange among agents. Based on recent research in this evolving field, multi-agent architectures can be categorized into four main types [35],[91],[94]. Additionally, we include alternative terminologies used in the literature to describe these categories.

2.1 Layered Structure (Hierarchical)

A layered or hierarchical structure organizes agents where each level supervises or controls the levels below it. Higher-level agents make strategic decisions, while lower-level agents perform specific tasks [37], [74]. This collaboration structure includes top-down control, where higher layers manage the lower layers for better decision-making. Complex tasks can be broken down to smaller and more manageable tasks [22]. The paper [23] introduces a self-collaboration code generation framework that uses LLMs as a multi agent team with assigning distinct roles that follows a waterfall model. Then, it divides the tasks into specialized roles: analyst, coder, and tester. This structure mirrors real-world software development hierarchies, where higher-level planning influences lower-level execution. This paper [122] follows a ChatGPT research group framework for a science experiment. The system is structured with seven AI assistants, which are defined as hierarchical workflow. Higher-level agents (strategic planning, literature search) guide and supervise the execution of research tasks by lower-level agents (data analysis, labware design, robotic operation).

2.2 Centralized Structure (Supervisor)

A single central agent (or controller) makes decisions for the entire multi-agent system. All other agents depend on this central entity for commands, coordination, and data sharing [33]. This framework makes the task assignment more efficient and enhances overall system reliability [107]. The authors of this paper [4] evaluate the power of LLM based multi agents to replicate human behaviors in experimental settings using a novel method named Turing Experiments. The system generates multiple human-like agents with different behaviors based on demographics that are

defined for them. These agents are supervised by a centralized agent like GPT-4 for decision-making and generating responses.

2.3 Decentralized Structure (Network)

In this architecture, agents operate independently or in small clusters without a single central controller [11],[61]. Decision-making is distributed, and agents interact directly with one another through peer-to-peer communication, enabling them to share information and coordinate actions dynamically [102],[13]. In a multi-agent debate framework designed by [107], multiple LLM based agents generate responses and engage in debates, and the final output is determined after multiple rounds of debate. RoCo [63] distributes decision-making across multiple robots, where each robot has its own LLM based agent, which independently communicates, plans, and adjusts its actions based on real-time dialogue. Robots engage in direct dialogue with each other instead of reporting to a central entity. This system is like CoELA (Cooperative Embodied Language Agent) framework [114], that enables each agent to operate independently using its own perception, memory, communication, planning, and execution modules.

2.4 Custom Structure

A custom structure is a flexible approach, where agents are organized dynamically based on the needs of the system [96]. It combines elements of layered, centralized, and decentralized structures, which allows agents to switch between these frameworks and can be designed to be application specific. The Generative Agent framework [69] is a custom multi-agent architecture to simulate believable, long-term human behavior in interactive environments. Besides each agent that operates independently with its own memory and reasoning module, a three-layered structure allows for continuous adaptation and emergent behavior. Both AgentVerse [16] and CAMEL [119] are custom multi-agent structures, while AgentVerse focuses on adaptive agent recruitment and structured collaboration for multi-agent problem-solving, CAMEL enables autonomous communication and cooperative task-solving using structured role-playing. Figure 1 provides a summary of these multi-agent structures.

Table 1 presents a comparison of various collaboration structures (e.g., hierarchical vs. decentralized), highlighting their respective strengths and weaknesses in terms of performance, computational cost, and applicability.

3 MULTI- AGENTS COLLABORATION STRATEGIES

The collaboration strategies refer to how agents exchange information to achieve coordinated decision-making and task completion to reach the shared or their individual goals [91]. This communication can be explicit, meaning that the agents directly send queries, responses, and other types of information to each other, or implicit when they are communicating through a real or simulated environment or structured prompts [19]. In this research paper we categorized these collaboration strategies into Foer main categories.

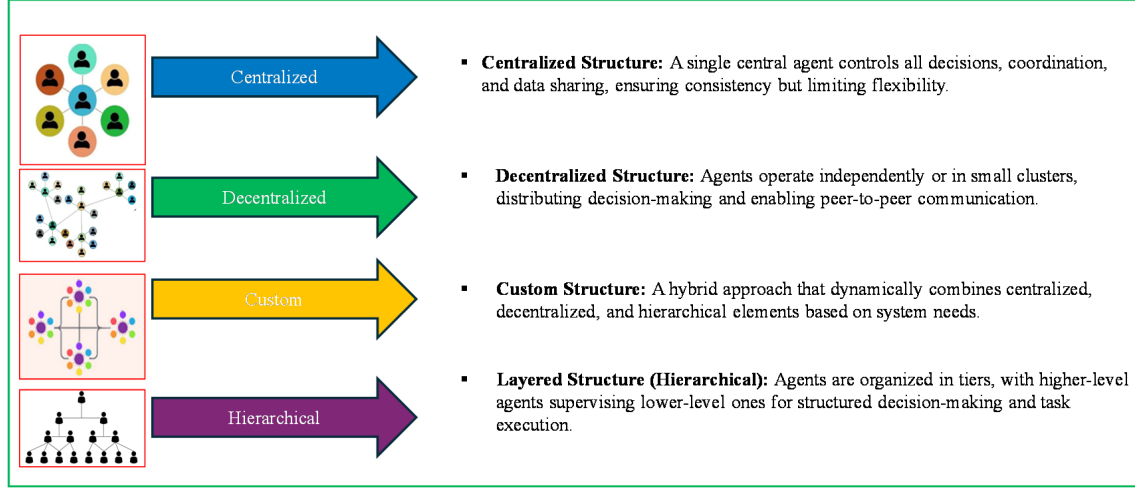


Figure 1: A Summary of Multi-Agent Collaboration Structure.

Table 1: Comparison of Collaboration Structures

Structure Type	Strengths	Weaknesses	Computing Cost	Performance	Applicability
Hierarchical (Layered)	Clear role allocation- Efficient task delegation- Scalable supervision	Bottlenecks at higher levels- Less flexible- Prone to failure at top nodes	Medium	High (if well-planned)	Software development, scientific research pipelines
Centralized (Supervisor)	Simpler coordination- Central decision-making improves control- Reduces redundancy	Single point of failure- Limits scalability- High dependency on central agent	Low to Medium	Moderate to High (efficient for small teams)	Social simulations, strategic decision-making
Decentralized (Networked)	Robust to failures- Scalable- Flexible peer-to-peer interactions	High coordination complexity- Harder to ensure consistency- Communication overhead	High	Variable (depends on agents' cooperation)	Robotics, sensor networks, multi-agent debates
Custom (Hybrid)	Adaptable to task/domain- Leverages benefits of other structures- Dynamic reconfiguration	Harder to design- May introduce complexity in decision-making and synchronization	Medium to High	High (if well-designed)	Simulations, economic modeling, adaptive systems

3.1 Cooperation

In this type of communication, agents work collaboratively towards a shared goal like document summarization [54], [95]. This collective intelligence can enhance the efficiency and problem-solving capabilities of agents [50]. However, there are some challenges regarding consistency in contribution and handling conflicts between agents. Several research papers used this method for multi agent design. For instance, Reflexion [84] shows a form of cooperation, where the actor, evaluator, and self-reflection models work together to improve the agent's decision-making through iterative learning. Each component provides valuable feedback that contributes to the overall improvement of the system. Agent4Rec is a generative agent for recommender systems [110] with cooperation-based MAS, but in an implicit and indirect manner. In this framework, agents work together to assess and interact with recommendation

systems to simulate user behavior. Unlike real-time cooperative systems like multi robots that pass information directly, Agent4Rec agents operate in parallel to test, refine, and evaluate recommendation models. Figure 2 provides an example of this multi-agent cooperation in medical research automation.

3.2 Competition

In this type, unlike cooperation, agents compete to achieve the best performance in each task. For example, multiple LLM agents may compete to generate the best legal argument [100]. This type of communication is useful when diverse solutions are needed, and it can enhance problem-solving and improve overall system performance. CompeteAI [120] is a competition-based multi-agent framework, designed to model and analyze competitive dynamics among LLM-driven agents. The framework simulates a virtual town, where

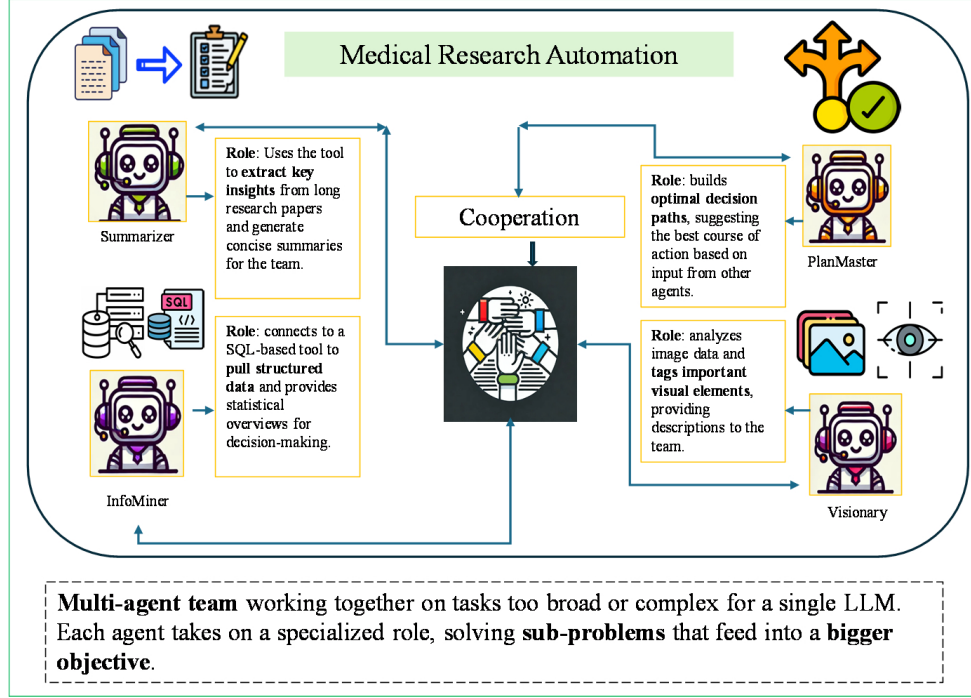


Figure 2: A Summary of Multi-agent Cooperation in Medical Research Automation.

restaurant agents compete for customers, analyzing strategies and market behaviors. GTBench [26] is a game-theoretic multi-agent framework, designed to evaluate LLMs’ strategic reasoning abilities through competition-based environments. It covers a wide range of game types and uses competing agents like GPT-4 and Llama-3 to test how well do LLMs perform against optimized search-based agents (e.g., Monte Carlo Tree Search, Tit-for-Tat strategy).

3.3 Debate

In a debate strategy, agents compete by arguing for and against a topic, helping to test ideas and find the truth. For example, in fact-checking misinformation, one agent supports a claim while others challenge it, making sure only accurate information is accepted [6]. Unlike general competition, which focuses on winning or being the best at a task, debate is about reasoning, persuasion, and comparing different viewpoints to make better decisions [112]. In the multi agent debate structure introduced by [57], the authors try to address the issue when LLMs become overconfident of their incorrect responses. This multi-agent architecture introduces debate-based interactions among multiple agents to encourage divergent thinking and enhance reasoning abilities. Similarly, FORD [89] has the framework simulates structured debates, where LLMs argue, counter-argue, and refine their reasoning to achieve consensus on commonsense reasoning tasks.

3.4 Hybrid Models

A hybrid multi-agent system combines elements of cooperation, competition, and debate, allowing agents to both refine and challenge each other’s outputs [29], [115]. For example, when we have

a mix of cooperation and competition, while agents try to get the advantage of knowledge-sharing between each other, they compete in some steps to enhance the reasoning through critiques [40]. This LLM based economic simulation framework [38] represents a hybrid of competition and cooperation, while agents compete for resources and employment as an individual goals, they follow shared economic rules, and their decisions collectively shape market stability. This reflects cooperative adaptation to societal norms.

4 MULTI- AGENTS LEARNING STRATEGIES

There are different strategies that LLM based MASs require to improve their performance, adapt themselves to a new environment, and better interact in the system. These learning strategies can be categorized based on how agents receive feedback, adjust their behaviors, and optimize their responses. In this paper, we focus on five of the most influential learning strategies commonly used in LLM based MAS. While the field is rapidly growing and new learning approaches may come overtime, these five categories currently represent the most widely recognized and impactful strategies.

4.1 Human-in-the-Loop Learning (HITL)

In this method of learning, agents rely on direct human feedback (e.g., rankings, corrections, annotations) to refine their results. Models adjust their behavior dynamically using labeled feedback or fine tuning to enhance response quality [37]. This ensures alignment with human values by explicitly incorporating human guidance. ClimB-DC [79], a human-guided and data-centric framework, integrates a multi-agent reasoning approach that systematically incorporates human feedback to enhance data processing. Domain

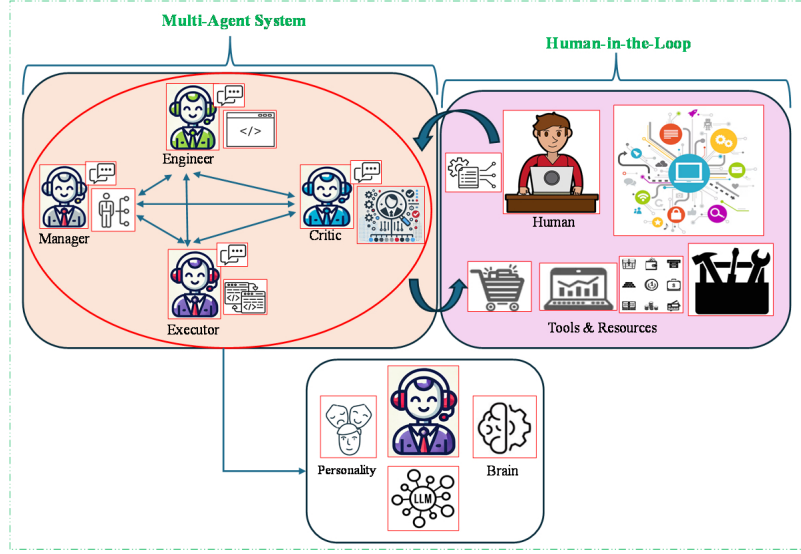


Figure 3: Human-in-the-Loop Learning process, where agents learn from human feedback, while utilizing tools and resources to improve their results.

experts (like a doctor analyzing medical records) guide the agent’s system by providing knowledge about the data or providing feedback in natural language, helping the system understand and process data correctly. Figure 3 illustrates that agents are capable of learning from human feedback and utilizing tools and resources to improve their performance.

4.2 Reinforcement Learning (RL)

In this learning strategy, agents refine their decision-making strategies using rewards and penalties they receive from an environment. RL enables LLM agents to optimize decision-making through trial and error, which is particularly useful in multi-agent coordination and competitive settings [52]. In Lyfe [46], one of the key elements is reinforcement learning (RL), which is integrated into the agents’ hierarchical action-selection system. Agents receive feedback from the environment (e.g., whether they receive useful information), and adjust their strategy accordingly. Similar to RL agents that learn from past experiences, Lyfe agents use a Summarize-and-Forget (SaF) memory system to ensure that agents remember only relevant experiences, improving their decision-making over time.

4.3 Prompt-Based Learning (Self-Refinement)

This is a lightweight and computationally efficient approach that uses prompt engineering techniques for more effective responses over time. Instead of retraining and fine-tuning the weights, agents iteratively refine their own prompts based on previous outputs. The GABSS system [99] integrates self-refining prompts using a thinking loop for each agent. By using this loop that includes daily planning, real-time decision-making and reflection, the system refines its prompts by updating agents’ memories and actions over time, instead of retraining the agents. CoA [119] divides a long-context task into chunks, and each worker agent processes a

segment and refines the response by summarizing and adding relevant details before passing it to the next agent. This way, instead of relying on a static prompt, each agent dynamically adjusts its prompt based on context from previous outputs.

4.4 LLM Fine-Tuning

In this method, LLMs that are representing agents, are fine-tuned using labeled datasets of expert decisions, and model weights are adjusted through supervised learning. This strategy is essential for domain-specific expertise, allowing LLMs to learn from high-quality expert demonstrations. In Trading GPT [55], the fine-tuning process involves training each LLM-based agent using multi-modal datasets (historical stock prices, trading records, financial news), and then adjusting model parameters based on past trading decisions and market events. This domain-specific fine-tuning improves decision-making and accuracy in financial trading. In this paper [34], fine-tuning occurs at multiple levels within the agent pipeline, with each agent specializing in a different stage of hallucination detection and mitigation. While Front-End agent creates a response based on user input, Second-Level reviewers and Third-Level reviewers are fine-tuned for hallucination detection and fact-checking improve the output.

4.5 Iterative Interaction Learning

This method allows agents to interact with themselves or other agents in an iterative loop (e.g., debates, strategic simulations), that can improve their problem-solving and decision-making. This allows agents to improve through simulations without external supervision. The Alympics framework [64] creates a multi-agent strategic simulation, where agents interact iteratively within a game-theoretic environment. The game involves 20-day survival competition where LLM-powered agents must bid for water resources daily, compete and collaborate dynamically as conditions

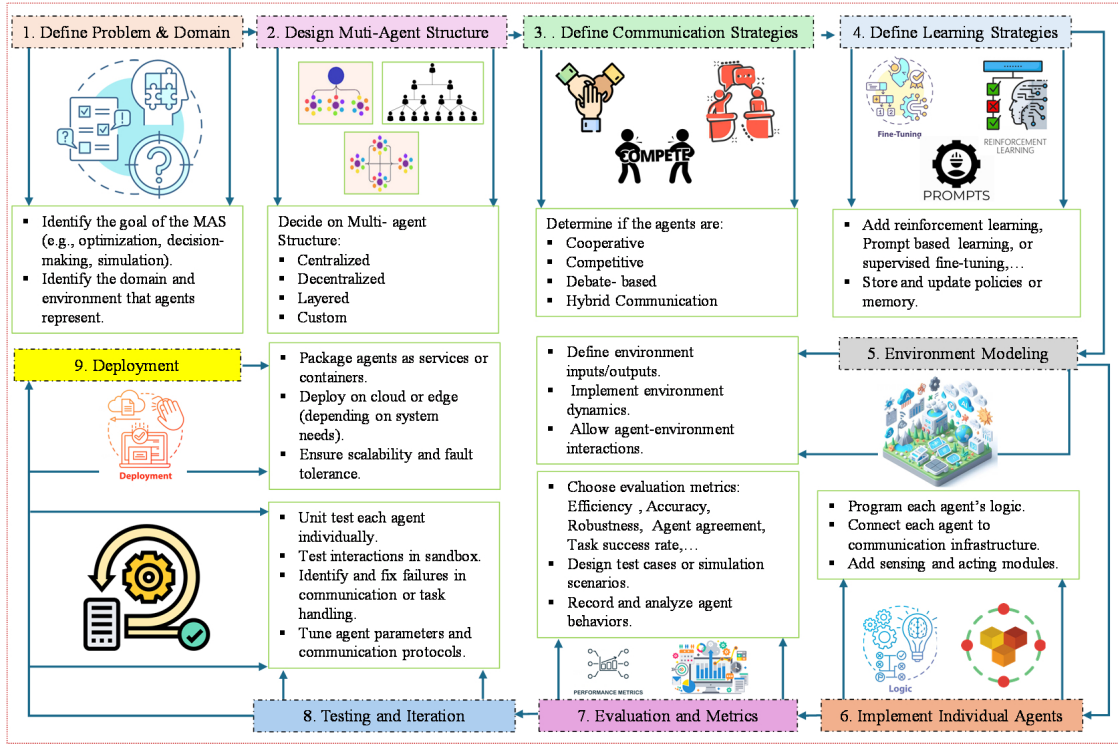


Figure 4: A Conceptual Roadmap for Designing a MAS based on LLMs.

change. Through repeated interactions, agents learn to refine their strategies without external supervision, which shows decision-making behaviors like humans. Similar to this study, Co-NavGPT [108] enables multiple LLM-powered robots to improve their path and exploration efficiency over repeated interactions and refine their selection strategies by interacting iteratively with each other through a global planner. In Appendix A, Tables 5 to 10, we summarize the current work according to their structure, collaboration strategies, and learning strategies.

4.6 Adaptation Challenges in MAS Learning

While LLM-specific techniques like prompt engineering and fine-tuning have become essential for enhancing agent specialization and contextual adaptation, their application in multi-agent systems (MAS) introduces new layers of complexity. In MAS, prompt engineering is often role-dependent—prompts are dynamically constructed or refined based on the agent’s role in a pipeline (e.g., planner, critic, executor) or the memory of previous interactions. This requires adaptive prompt generation strategies that can accommodate inter-agent communication and evolving context, as seen in frameworks like CAMEL [53] and GABSS [99].

Fine-tuning, on the other hand, enables domain-specific knowledge to be embedded into each agent, allowing for diverse expertise across a system. However, applying fine-tuning in MAS faces several challenges:

- **Scalability:** Fine-tuning each agent separately may lead to large resource consumption.

- **Catastrophic Forgetting:** When agents undergo sequential fine-tuning (e.g., when adapting to new tasks), they risk losing previously learned behaviors or decision strategies. This becomes critical in MAS where agents must retain shared coordination norms or negotiation histories across interactions.
- **Consistency Maintenance:** In multi-agent settings, inconsistent fine-tuning can lead to misalignment between agents, undermining cooperation and decision-making.

To address these issues, recent research explores hybrid approaches, such as combining lightweight prompt tuning with shared memory modules or using continual learning techniques that maintain historical task knowledge while integrating new information [101], [98].

5 MULTI- AGENTS METRICS EVALUATION

There is different evaluation metrics based on whether the system is a single-agents or multi-agents’ system. It can also vary by domain, in which the system is defined, such as Social Science, Natural Science, and Engineering. Below we review three major domains and list some of the important evaluation metrics that have been used for both single-agents and multi-agents.

5.1 Social Science

Applications in social science include decision-making, social interactions, ethics, and policy making. The evaluation metrics are designed to capture both individual and collaborative aspects of

the MASs. Single-agent metrics typically focus on language quality [36], fairness, ethical considerations, and user engagement [78]. For example, social chat bots are commonly evaluated based on their coherence, engagement, and potential bias in responses. In contrast, multi-agent metrics emphasize the dynamics of interaction, such as social negotiation, cooperation, and consensus-building. MASs used in negotiation tasks are often assessed using metrics like agreement rate and persuasion success, while AI-driven policymaking simulations rely on fairness and consensus metrics to improve the quality and inclusiveness of generated policies. Appendix A, Table 2, summarizes some of these metrics for single-agent and multi-agent systems.

5.2 Natural Science

In the Natural Science domain, single-agent metrics emphasize accuracy, factual correctness, and generalization, while multi-agent metrics focus on collaborative knowledge discovery, hypothesis validation, and consensus building. For example, single-agent systems are evaluated on fact consistency and citation validity, whereas multi-agent systems are assessed using agreement rate, contradiction detection, logical consistency, and computational efficiency. Appendix A, Table 3, summarizes some of these metrics for single and multi-agents LLMs.

5.3 Engineering

In the Engineering domain, single-agent metrics focus on code correctness, performance, and execution efficiency [25], often applied to AI coding tools like Copilot or CodeLlama [73] [67], using metrics such as Pass@k and Compilation Success [20]. Multi-agent metrics assess collaborative software engineering, distributed problem solving, and system integration. An example is MetaGPT [37], which is evaluated on Module Integration, Task Distribution, and Task Synchronization in robotics applications. Appendix A, Table 4, summarizes selected metrics for single-agent and multi-agent approaches in this domain.

5.4 Domain Specific Metrics

In domain-specific applications, specialized metrics provide deeper insights into system efficacy. For example, in robotics-based MAS, metrics like task completion rate, navigation success, and mean time to task completion are widely used to evaluate agent coordination and task efficiency, as in the collaborative robot frameworks presented by [31], [74]. In debate-based multi-agent systems, such as LLM-driven deliberation models, consensus quality, argument diversity, and factual consistency are employed to assess discourse efficacy and epistemic alignment [101], [81]. In healthcare MAS, systems are evaluated through metrics such as diagnostic agreement rate, treatment recommendation validity, and false-positive reduction — essential in ensemble medical diagnosis agents [65], [94]. In finance, especially in systems like credit score improvement, metrics such as goal alignment score, plan feasibility rate, and reduction in invalid recommendations are critical to assessing user-centric recommendation quality and regulatory robustness [93].

6 LIMITATIONS & OPEN PROBLEMS

The rise of LLM-based MAS is shifting artificial intelligence to a new era. These systems allow agents to work together, discuss, or compete in different ways, more like human interactions. Our review of the literature shows that there is great potential that MAS can solve complex problems, learn on their own, and make decisions without a central leader [120]. LLM-powered MASs leverage communication strategies such as competition, cooperation, and debate. They enable flexible interactions, including hierarchical, centralized, and decentralized collaboration structures, which show their intelligence and problem-solving capabilities [84], [110]. However, there are still some challenges and limitations that need to be addressed in these systems. Several limitations persist in current MAS implementations:

6.1 Coordination Complexity

As the number of agents increases, effective communication and task allocation become more challenging, especially in decentralized multi-agent environments, where LLM based agents need to synchronize their plans and actions efficiently and there is no leader [114]. This problem becomes more challenging when agents are assigned to various roles or their tasks change dynamically. This emphasizes the need for better methods to help agents coordinate smoothly and solve conflicts as the system grows larger.

6.2 Computational and Energy Efficiency

There are growing concerns about the energy consumption and computational resources required by MAS. To improve time efficiency, most architectures emphasize parallel execution, allowing agents to run simultaneously and process tasks more quickly. However, this approach can further increase the demand for energy and processing power [51]. As the number and complexity of agents increases, the demand for powerful hardware and memory resources becomes a major problem, which limits scalability and practical deployment. High energy consumption has significant environmental impacts, making it crucial to explore energy efficient models, and optimized training pipelines [86].

6.3 Alignment and Safety

Another open challenge is to make these MASs to follow human ethics and values and behave safely. These agents are prone to show unexpected behaviors, such as generating incorrect information (hallucinations) or spreading biased content, which can make the system less reliable [81]. Due to the MAS communications, a mistake made by one agent can quickly affect the decisions of others, making the problem even worse. To solve this, we need better ways to guide agents to act safe and fair, and methods to test whether they are working as intended [28].

6.4 Explainability & Trustworthiness

Understanding why agents make certain decisions, particularly in complex or unexpected situations, is difficult. The lack of explainability reduces the trust in MAS and makes it hard for the user to rely on agents' recommendations and decisions. Improving interpretability is important for building user confidence and making

sure that MAS can be safely used in sensitive areas like healthcare, finance, or autonomous driving [109], [1].

7 Conclusion

In this review paper, we studied the structured analysis of LLM based MAS, which was focused on their communication strategies, learning mechanisms and collaboration structures. Our findings show that well-designed agent frameworks can make teamwork more effective, and results in better decision-making and adaptive learning. This study also emphasizes how MAS can leverage the strengths of multiple specialized agents to enhance efficiency and problem-solving capabilities. There are some open problems for MAS that require further work and investigations. The issues related to fairness, bias and ethical considerations in the interactions of agents remain open problems. Despite significant advancements, MAS still faces challenges such as high computational costs, coordination complexity, and the need for more efficient learning techniques. Future research should focus on improving the scalability and efficiency of MAS and developing advanced learning algorithms for better adaptability. This will result in enhancing real-world deployments across industries. Additionally, addressing ethical concerns, ensuring fairness in agent interactions, and optimizing computational efficiency will be crucial for the widespread adoption of these systems. Further interdisciplinary collaboration between AI, robotics, and cognitive science can lead to more robust and human-like multi-agent intelligence.

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APPENDIX A

Table 2: Evaluation Metrics in the Social Science Domain with Example Publications.

Domain	Metric Name	Agent Type	Short Explanation	Example Publication
Social Science	Coherence	Single-Agent	Measures logical consistency in chat bot responses	[8]
Social Science	Engagement	Single-Agent	Assesses user interest during interaction	[83]
Social Science	Bias	Single-Agent	Detects unfair or prejudiced language	[5]
Social Science	Agreement Rate	Multi-Agent	Tracks frequency of mutual decisions	[3]
Social Science	Persuasion Success	Multi-Agent	Measures of success in convincing peers	[7]
Social Science	Fairness Metrics	Multi-Agent	Evaluates equity in outcomes or decisions	[44]
Social Science	Consensus Metrics	Multi-Agent	Assesses group alignment on decisions	[14]

Table 3: Evaluation Metrics in the Natural Science Domain with Example Publications.

Domain	Metric Name	Agent Type	Short Explanation	Example Publication
Natural Science	Accuracy	Single-Agent	Measures correctness of generated predictions or outputs	[123]
Natural Science	Factual Consistency	Single-Agent	Checks accuracy of generated scientific information	[31]
Natural Science	Generalization	Single-Agent	Evaluates the model’s ability to apply knowledge to unseen scenarios	[98]
Natural Science	Citation Validity	Single-Agent	Assesses the correctness and relevance of cited information	[10]
Natural Science	Agreement Rate	Multi-Agent	Measures consistency in jointly generated hypotheses	[88]
Natural Science	Contradiction Detection	Multi-Agent	Identifies logical inconsistencies across agent outputs	[87]
Natural Science	Logical Consistency	Multi-Agent	Assesses internal coherence of multi-agent reasoning	[62]
Natural Science	Computational Efficiency	Multi-Agent	Evaluates resource use during collaborative analysis	[17]

Table 4: Evaluation Metrics in the Engineering Domain with Example Publications.

Domain	Metric Name	Agent Type	Short Explanation	Example Publication
Engineering	Pass@k	Single-Agent	Measures the success rate of code generation tasks within k attempts	[92]
Engineering	Compilation Success	Single-Agent	Assesses whether the generated code compiles without errors	[45]
Engineering	Module Integration	Multi-Agent	Evaluates the effectiveness of integrating independently developed modules	[48]
Engineering	Task Distribution	Multi-Agent	Measures the efficiency of task allocation among multiple agents	[77]
Engineering	Task Synchronization Score	Multi-Agent	Assesses the coordination and timing accuracy among collaborating agents	[49]

Table 5: Summary of the LLM-MA studies. We categorize current work according to their structure, collaboration strategies, and learning strategies. For learning strategies, we used abbreviations: Human in the loop (HITL), Prompt based learning (PBL), Reinforcement Learning (RL), LLM fine tuning (FT), and Iterative Interaction Learning (IIL).

Paper	Structure	Collaboration Strategy	Learning Strategy	Year
[37]	Layered	Hybrid	HITL, RL, FT	2023
[63]	Decentralized	Cooperation	PBL, HITL	2024
[69]	Custom	Hybrid	PBL, IIL	2023
[99]	Custom	Hybrid	PBL	2023
[103]	Custom	Hybrid	RL, IIL	2023
[64]	Custom	Hybrid	IIL	2023
[55]	Custom	Hybrid	RL, FT	2023
[24]	Decentralized	Debate	IIL, PBL	2024
[11]	Decentralized	Debate	IIL, PBL	2023
[61]	Decentralized	Hybrid	IIL, RL	2023

Table 6: Summary of the LLM-MA studies. We categorize current work according to their structure, collaboration strategies, and learning strategies. For learning strategies, we used abbreviations: Human in the loop (HITL), Prompt based learning (PBL), Reinforcement Learning (RL), LLM fine tuning (FT), and Iterative Interaction Learning (IIL).

Paper	Structure	Collaboration Strategy	Learning Strategy	Year
[66]	Custom	Hybrid	HITL, PBL, IIL	2024
[111]	Custom	Cooperation	HITL, PBL, IIL, FT	2024
[93]	Custom	Hybrid	HITL, RL, FT	2023
[12]	Custom	Cooperation	IIL, PBL	2023
[15]	Custom	Cooperation	IIL, PBL	2024
[53]	Custom	Cooperation	IIL, PBL, HITL	2023
[23]	Layered	Cooperation	IIL, PBL	2024
[118]	Layered	Cooperation	IIL, PBL	2024
[74]	Layered	Cooperation	Hybrid	2024
[42]	Layered	Debate	Hybrid	2023

Table 7: Summary of the LLM-MA studies. We categorize current work according to their structure, collaboration strategies, and learning strategies. For learning strategies, we used abbreviations: Human in the loop (HITL), Prompt based learning (PBL), Reinforcement Learning (RL), LLM fine tuning (FT), and Iterative Interaction Learning (IIL).

Paper	Structure	Collaboration Strategy	Learning Strategy	Year
[22]	Layered	Hybrid	RL	2023
[114]	Decentralized	Cooperation	HITL, PBL, IIL, FT	2023
[18]	Custom	Cooperation	IIL, PBL	2024
[107]	Centralized	Cooperation	IIL, PBL	2023
[13]	Decentralized	Cooperation	IIL, PBL	2023
[122]	Layered	Cooperation	HITL, RL, PBL	2023
[101]	Custom	Debate	HITL, PBL, IIL, FT	2023
[89]	Custom	Cooperation	HITL, PBL, IIL	2023
[70]	Custom	Hybrid	IIL, PBL	2023
[29]	Custom	Hybrid	HITL, PBL, IIL, FT	2023

Table 8: Summary of the LLM-MA studies. We categorize current work according to their structure, collaboration strategies, and learning strategies. For learning strategies, we used abbreviations: Human in the loop (HITL), Prompt based learning (PBL), Reinforcement Learning (RL), LLM fine tuning (FT), and Iterative Interaction Learning (IIL).

Paper	Structure	Collaboration Strategy	Learning Strategy	Year
[46]	Custom	Hybrid	RL	2023
[52]	Custom	Hybrid	RL	2023
[102]	Decentralized	Hybrid	IIL, RL	2023
[33]	Centered	Cooperation	IIL, RL	2023
[59]	Decentralized	Hybrid	IIL, RL	2023
[65]	Decentralized	Hybrid	IIL, RL	2023
[54]	Decentralized	Cooperation	IIL, RL	2023
[27]	Decentralized	Hybrid	IIL, RL	2024
[50]	Decentralized	Cooperation	IIL, RL	2024
[116]	Custom	Hybrid	IIL, RL	2023

Table 9: Summary of the LLM-MA studies. We categorize current work according to their structure, collaboration strategies, and learning strategies. For learning strategies, we used abbreviations: Human in the loop (HITL), Prompt based learning (PBL), Reinforcement Learning (RL), LLM fine tuning (FT), and Iterative Interaction Learning (IIL).

Paper	Structure	Collaboration Strategy	Learning Strategy	Year
[39]	Decentralized	Hybrid	IIL, RL	2023
[120]	Decentralized	Competition	IIL, RL	2023
[110]	Decentralized	Cooperation	IIL, RL	2024
[117]	Custom	Cooperation	IIL, RL	2024
[99]	Custom	Cooperation	IIL, RL	2023
[41]	Custom	Hybrid	IIL, RL	2023
[95]	Decentralized	Cooperation	IIL, RL	2023
[32]	Custom	Cooperation	IIL, RL	2023
[96]	Custom	Cooperation	IIL, RL	2023
[84]	Custom	Cooperation	IIL, RL	2024

Table 10: Summary of the LLM-MA studies. We categorize current work according to their structure, collaboration strategies, and learning strategies. For learning strategies, we used abbreviations: Human in the loop (HITL), Prompt based learning (PBL), Reinforcement Learning (RL), LLM fine tuning (FT), and Iterative Interaction Learning (IIL).

Paper	Structure	Collaboration Strategy	Learning Strategy	Year
[58]	Custom	Debate	IIL	2023
[4]	Centralized	Cooperation	IIL, RL	2023
[26]	Decentralized	Competition	IIL	2024
[100]	Layered	Competition	RL	2024
[113]	Layered	Debate	IIL	2024
[6]	Layered	Debate	IIL	2024
[99]	Custom	Hybrid	RL	2023
[80]	Custom	Cooperation	HITL	2025
[60]	Custom	Cooperation	HITL	2025
[34]	Layered	Cooperation	FT	2025